

Single cycle control (Note)

- Control Signal.

- In a single cycle processor, control is managed by a dedicated control unit that generates signals based on the current instruction.

- Control unit checks the specific field in the 32 bit instruction fetched from the memory and produce control signal.

eg: Consider R type instruction.

Instruction Format

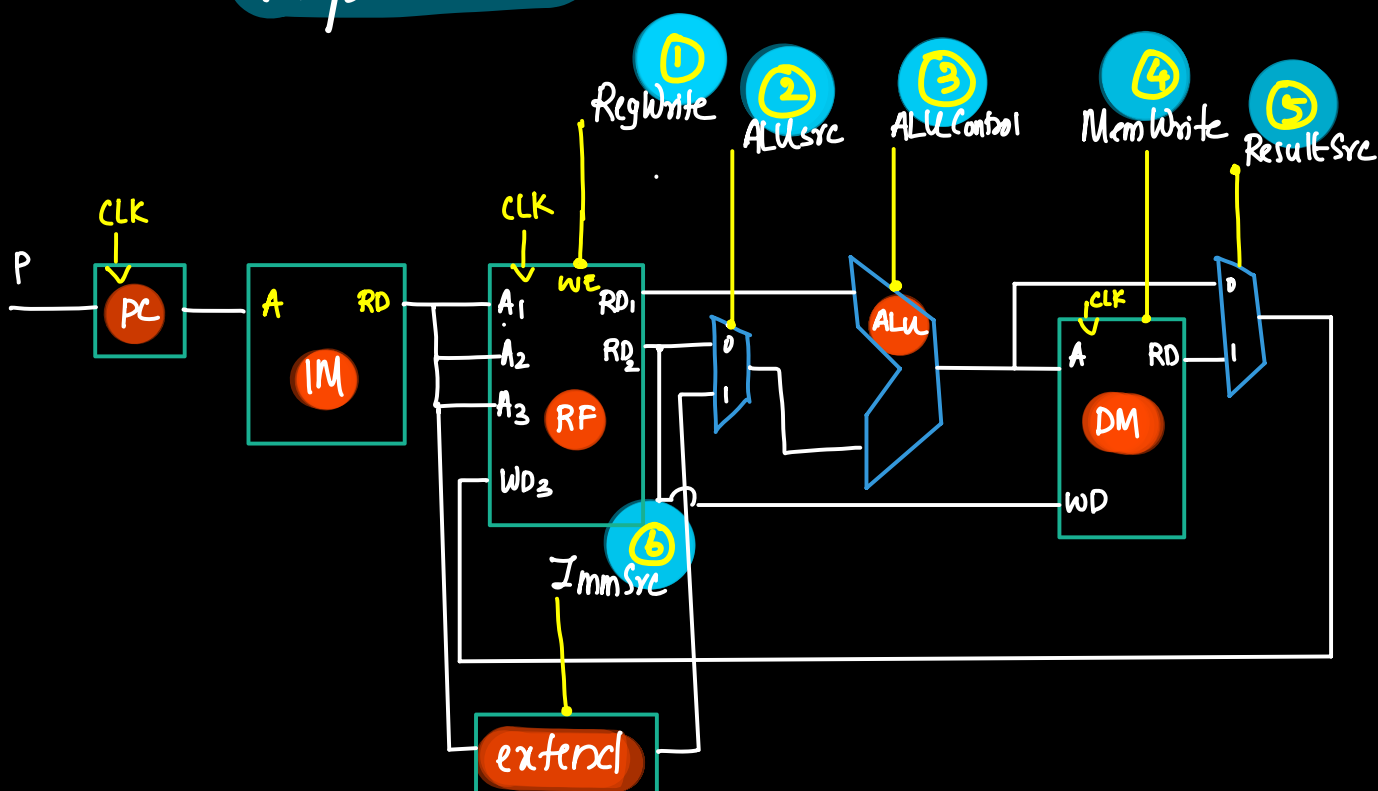
7	5	5	3	5	7
funct7	rs1	rs2	funct3	rd	op



- Input to the control units are

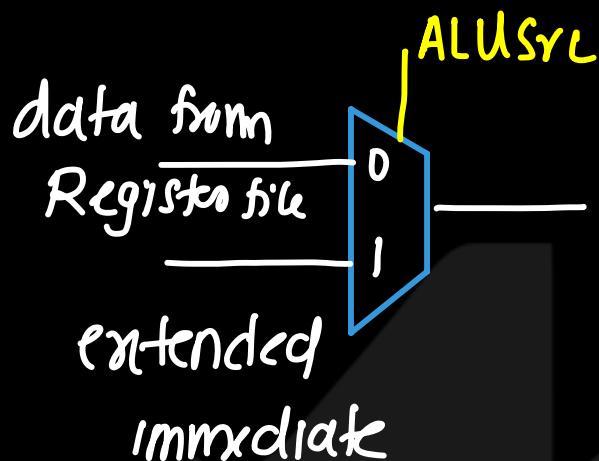
- ① op (opcode field)
- ② funct3 field
- ③ funct7 field

Important Control signals.



① RegWrite (Register write) (N)
→ this signal is enabled when instruction needs to write the result back to register file
eg: If RegWrite = 1, then data written back to the register file

② ALUSrc
It is a multiplexer select line that determines if the second input of ALU comes from register file or from extended immediate value.



If ALUSrc = 0, then 2nd input of ALU is from register file

If ALUSrc = 1, then 2nd input of ALU is from extended immediate

③ ALU control

→ this specifies the operation need to perform
eg: For addition (000)
For sub (001) etc

④ MemWrite (Memory Write).

→ this signal is enabled when we need to write data into the data memory.

⑤ ResultSrc

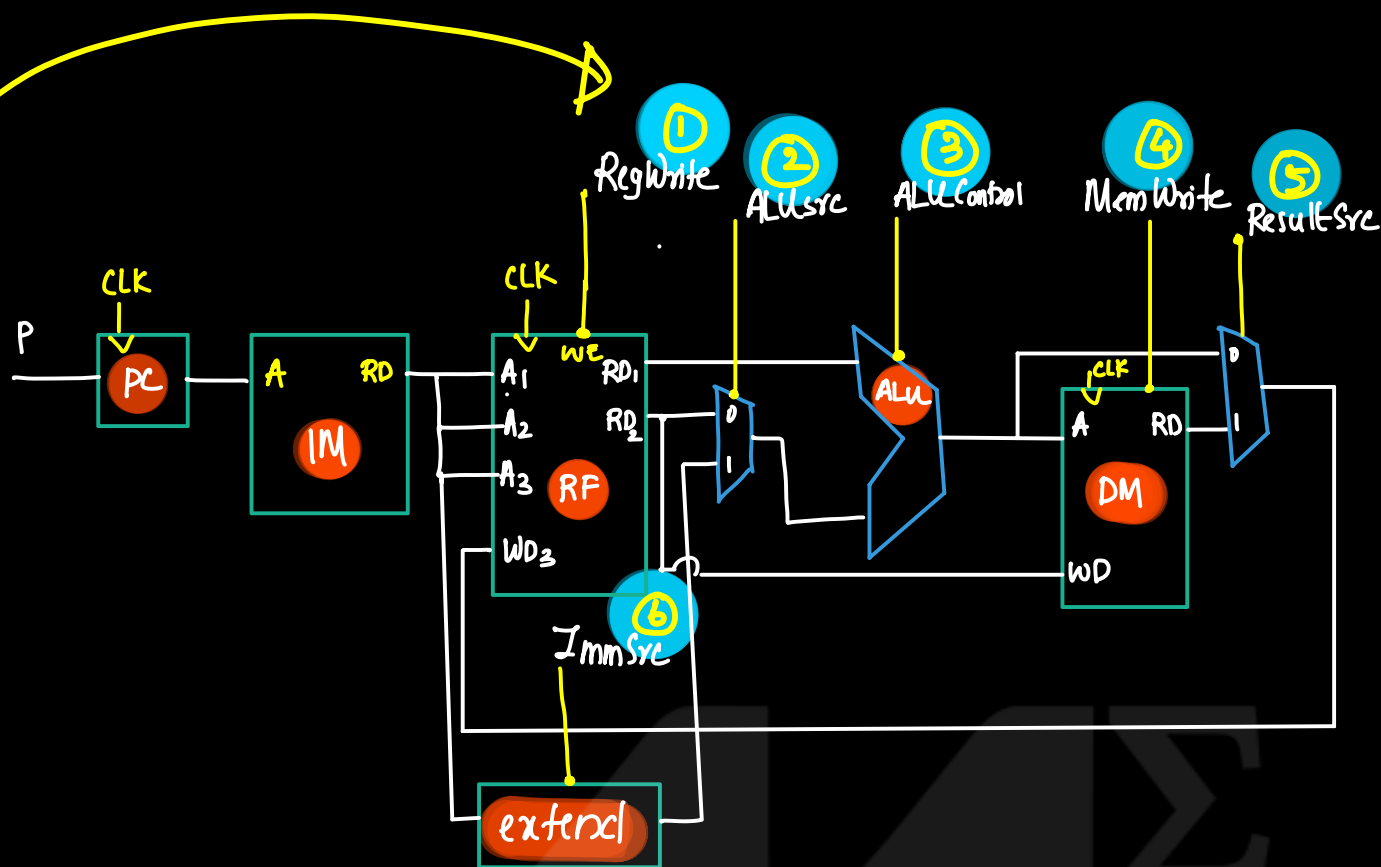
— it is a multiplexer select signal
→ this signal used to select the source of data to be stored either ALU result or Data memory output.

⑥ PCSrc

this signal determines the next program counter value.

Control unit for single cycle RISC processor consists of two parts

- ① Main decoder
- ② ALU decoder



① Main decoder

- main decoder determines the instruction type (I, R, etc.) from the opcode and produce the appropriate signal.
- Control signal produced by the main decoder

① ② ④ ⑤ ⑥

- ① RegWrite
- ② ALUSrc
- ③ MemWrite
- ④ ResultSrc
- ⑤ ImmSrc

- Internal Control signal produced by the main decoder
 - Branch
 - ALUop

② ALU decoder

②

— ALU decoder generate control signal to specify the ALU operation need to perform.

— Input to the ALU decoder

(1) ALUop (from the main decoder)

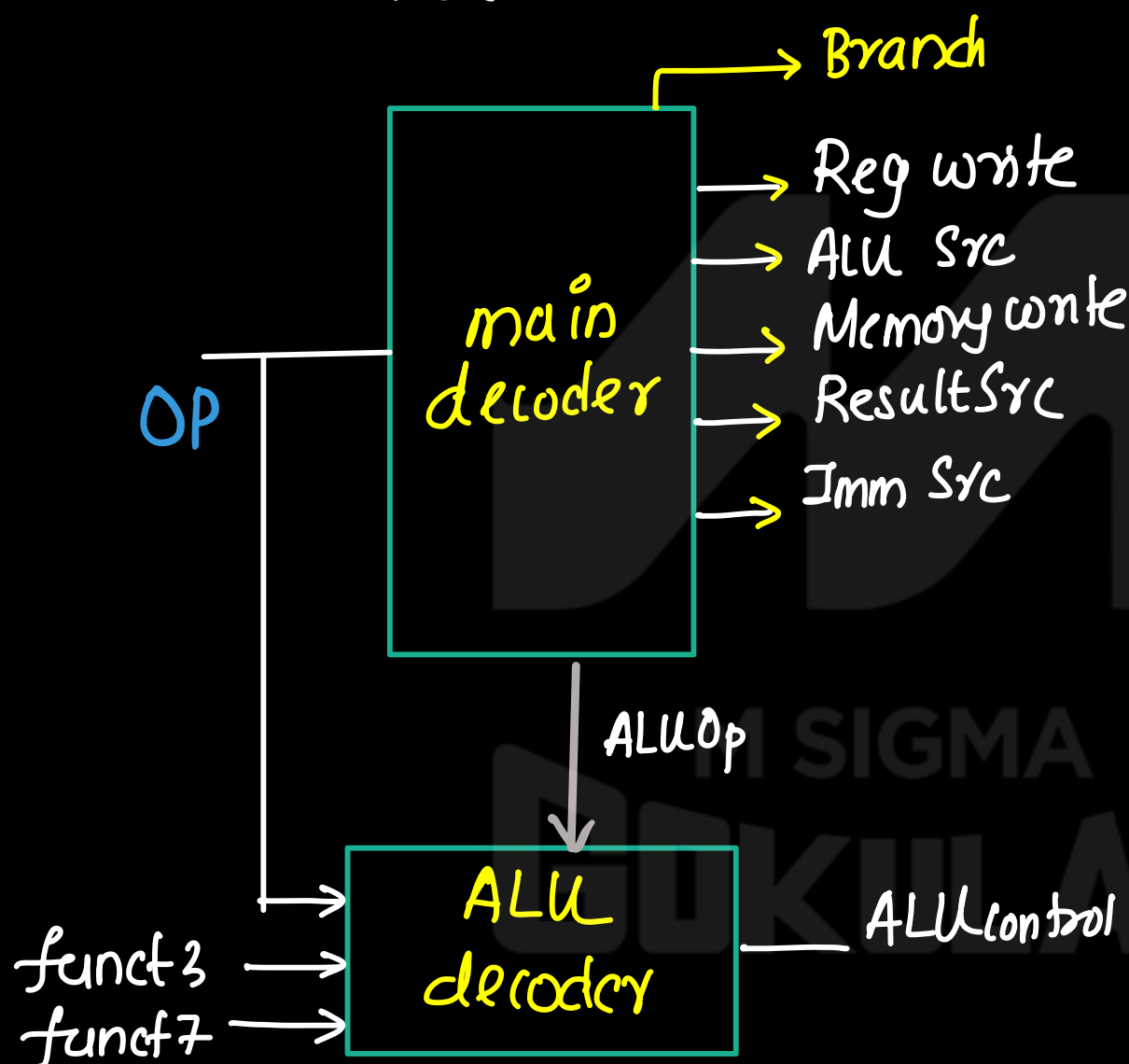
(2) funct3

(3) funct7

(4) op field

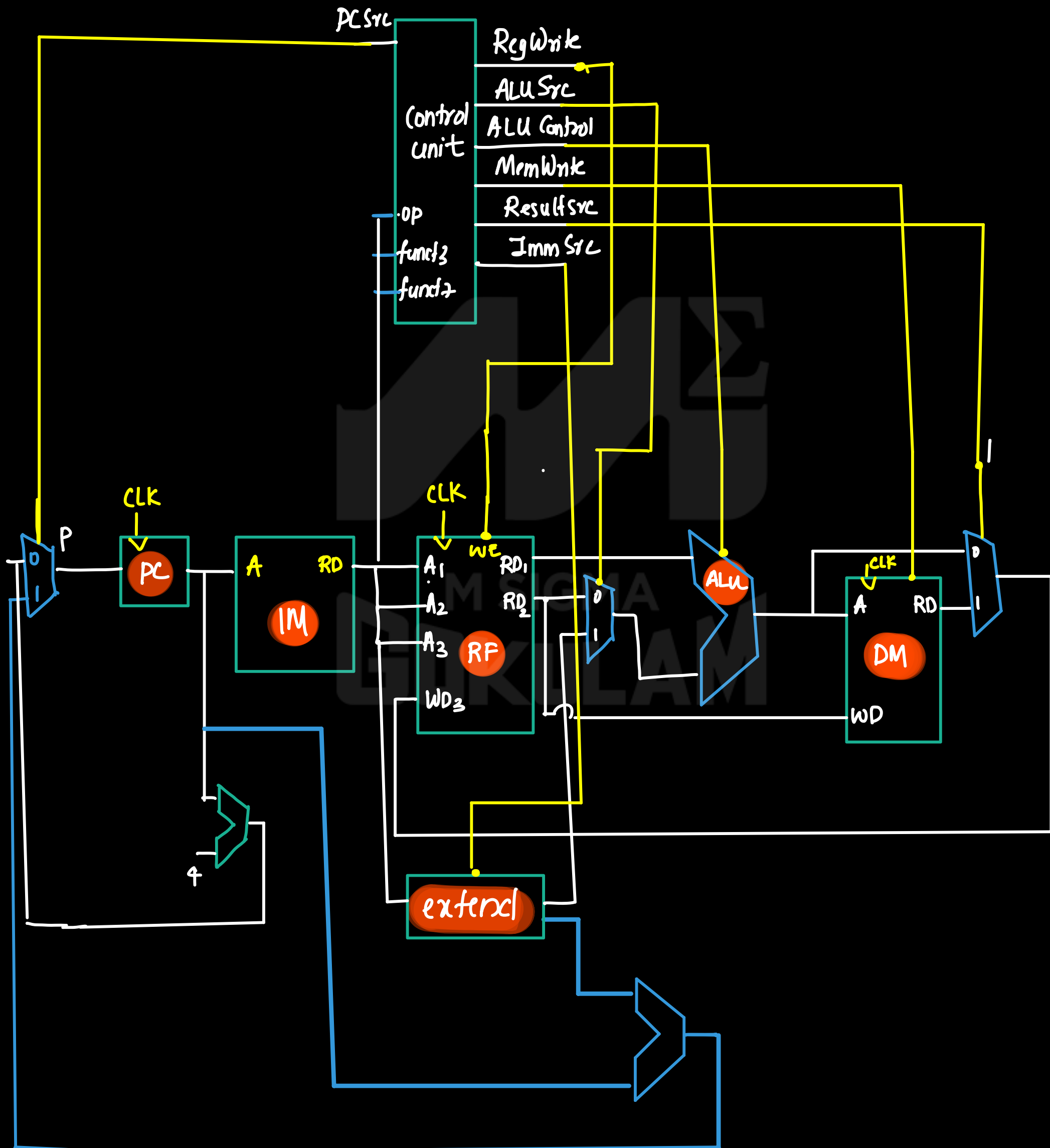
— Output

ALU control signal.



Q. Draw the single cycle data path diagram with control signals. (2)

Complete single cycle processor
(Data path and control path)



Single cycle processor Performance (Note)

— Program execution time (T_{ex})

$$T_{ex} = \text{no. of instructions executed} \times \text{cycles per instruction} \times \text{clock cycle time}$$

$$T_{ex} = \text{instruction count (IC)} \times \text{CPI} \times T_{\text{cycle}}$$

$$T_{ex} = IC \times CPI \times T_{\text{cycle}}$$

CPI for single cycle processor

In a single cycle processor, every instruction execute in single clock cycle. So $CPI = 1$

$$T_{ex} = 1 \times T_{\text{cycle}}$$

this means that performance depends mainly on clock period

Cycle time.

— In single cycle processor every instruction execute in one cycle. So clock period must be long for the slowest instruction.

— Single cycle RISC processor slowest instruction is load instruction (lw)

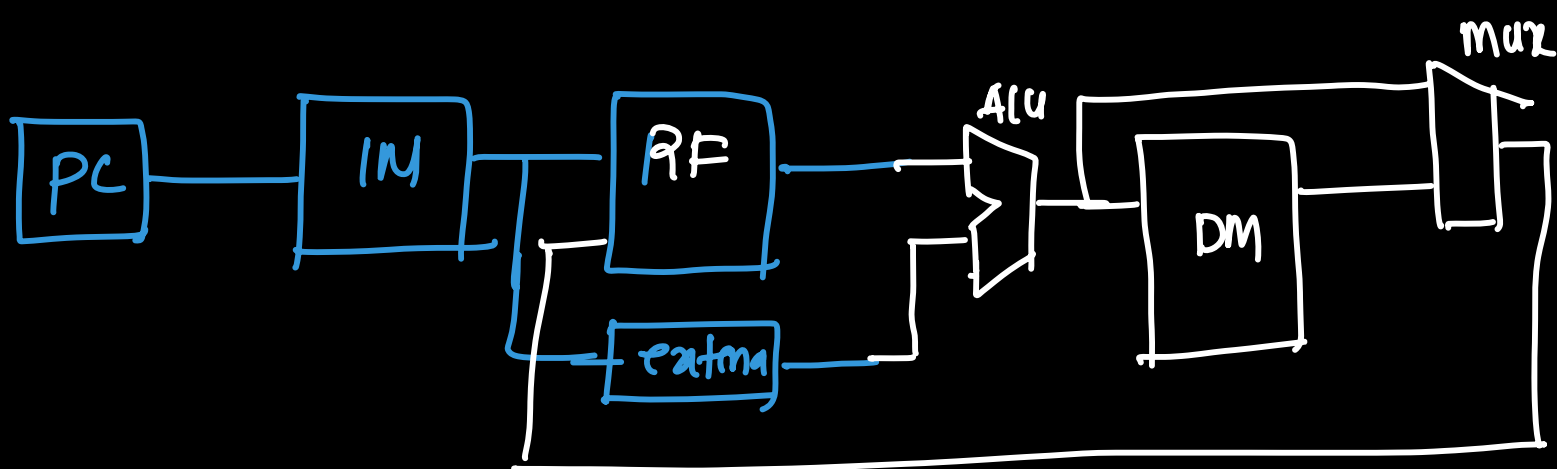
eg: `lw x5, 4(x9)`

$x9 \rightarrow 2000$

$4 + 2000 \rightarrow 2004 \rightarrow \text{memory location}$

Let content of 2004 is 06.

→ load content of memory location 2004 in the register $x5$



- Steps :
- ① PC give the address of instruction to be executed
Time taken by the PC to output address — t_{PC}
 - ② Instruction fetch from the instruction memory.
Time for IM to output the instruction bits after receiving the address from PC — t_{IM}
 - ③ Read the register file. (read the content of s_r)
Time to read the register file ($T_{RF(read)}$)
 - ④ Generate 32 bit value from the immediate field in the instruction bits (4 $\rightarrow$ 32 bit value)
Time to extract immediate field and convert it to full width signed value. (t_{imm})
 - ⑤ ALU perform the addition.
Time required for producing output (t_{ALU})
 - ⑥ Read the content from data memory.
Time for read the data memory (t_{DM})
 - ⑦ Mux select Data memory output.
Time for multiplexer to select one (t_{mux})
 - ⑧ output of data memory given to the write data input of Register file
Time the data must be ready at the register file input so that register can store it correctly ($t_{RF(setup)}$)

$$T_{cycle} = t_{PC} + t_{IM} + t_{RF(read)} + t_{imm} + t_{ALU} + t_{DM} + t_{mux} + t_{RF(setup)}$$

compute execution time for the program with 100 billion instruction



Table 7.7 Delay of circuit elements

Element	Parameter	Delay (ps)
Register clk-to-Q	t_{pcq}	40
Register setup	t_{setup}	50
Multiplexer	t_{mux}	30
AND-OR gate	t_{AND-OR}	20
ALU	t_{ALU}	120
Decoder (control unit)	t_{dec}	25
Extend unit	t_{ext}	35
Memory read	t_{mem}	200
Register file read	t_{RFread}	100
Register file setup	$t_{RFsetup}$	60

$T_{ex} = IC \times CPI \times T_{cycle}$

$IC = 100 \times 10^9$

$CPI = 1$

$T_{cycle} = 4$

